Master 2 BDEEM

Marketing Data Analysis

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Materials on Moodle

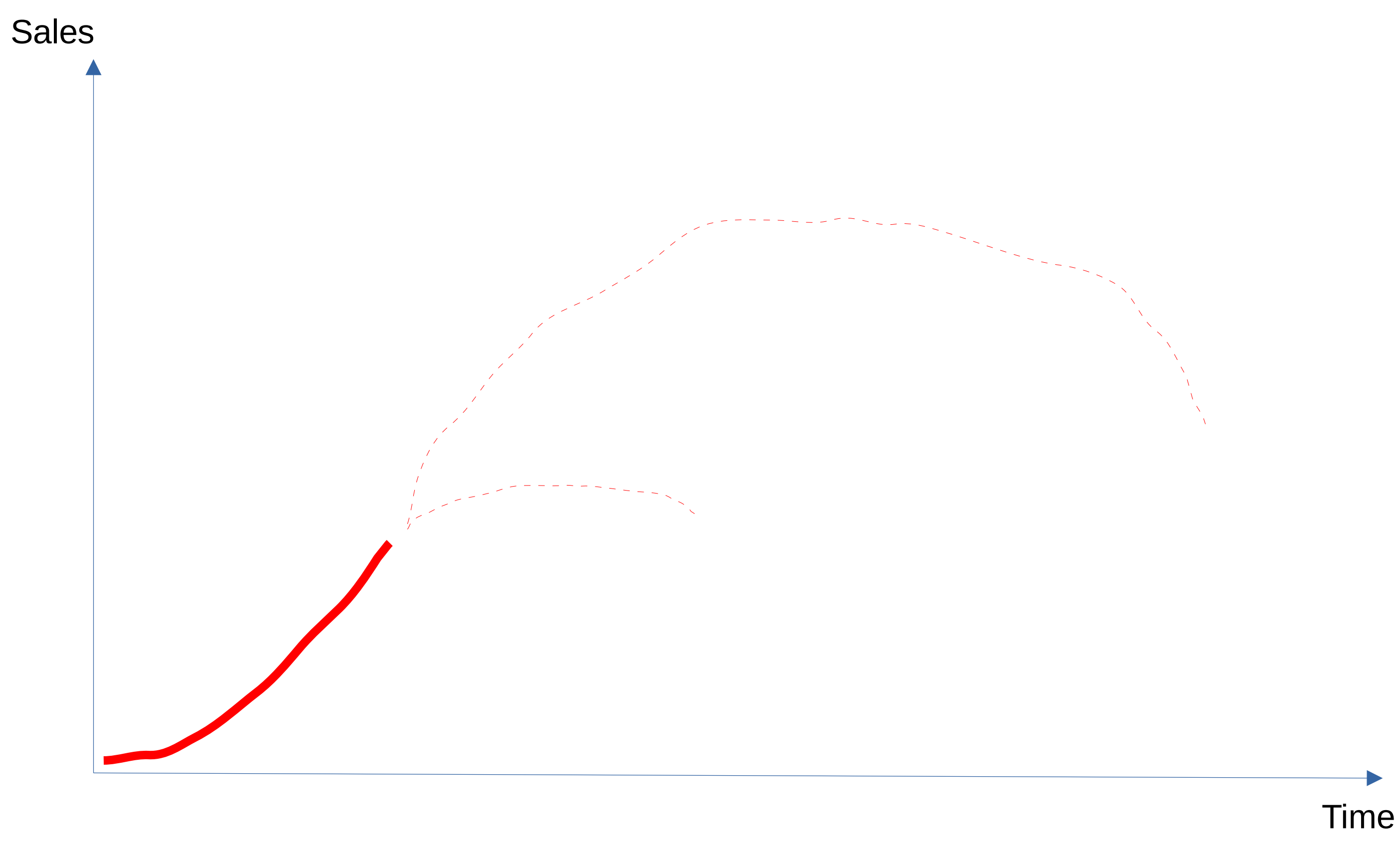
Course Description

1) Diffusion Models and the Bass Model for effective marketing strategies (linear regression).
2) Conjoint Analysis for a better understanding of consumer preferences and choices to meet their needs and desires (Regression).
3) Segmentation to group consumers based on shared characteristics, enabling more personalized marketing strategies (Hierarchical agglomerative clustering).
4) Perceptual Mapping and Positioning to understand the positions of products in the minds of consumers (Principal component analysis)

Aggregate Demand Estimation
The BASS Model

Adoption of a new product,
Diffusion Model
and Product Life Cycle

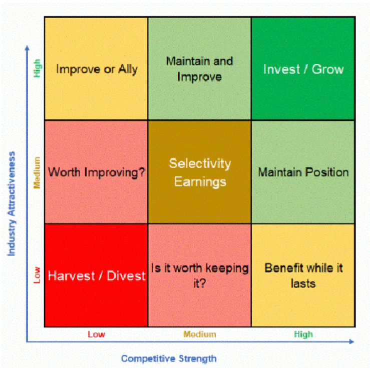
The product life cycle

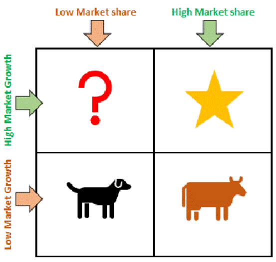


Why anticipate the evolution of the market size ?

- Financial Planning
- Production Planning
- Customer Satisfaction

- Resource Allocation & Strategic Planning
  - to manage the business portfolio of the company
  - to direct the growth of the company towards the most interesting Strategic Business Unit (SBU)





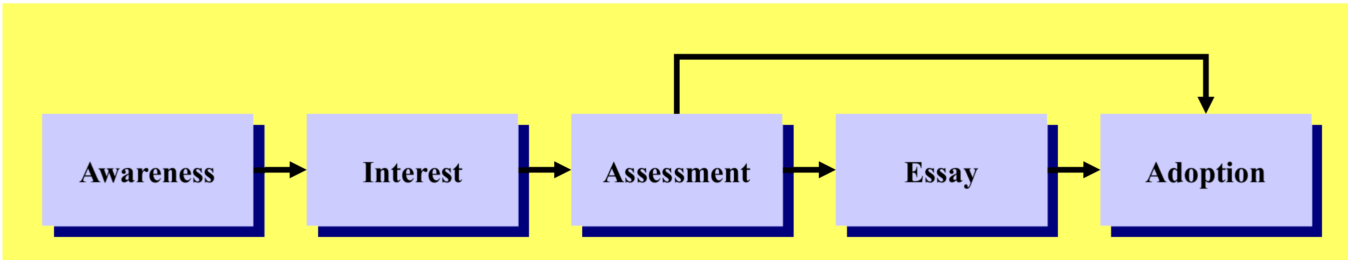
How anticipate the diffusion of a new product ?

- Market Research
- Competitor Analysis
- Pilot Testing
- Continuous Monitoring
- Early Adopter Identification
- Predictive Analytics

Adoption Process

The adoption process is called

- the mechanism through which potential customers become aware of a new product, sometimes try it, and ultimately adopt (or reject) it
- the mental pattern followed by an individual from the first information he receives about an innovation until the moment when he adopts it definitively



Diffusion Process

- The process of diffusion is the mechanism through which the innovation by its adoption enters or spreads within the market.

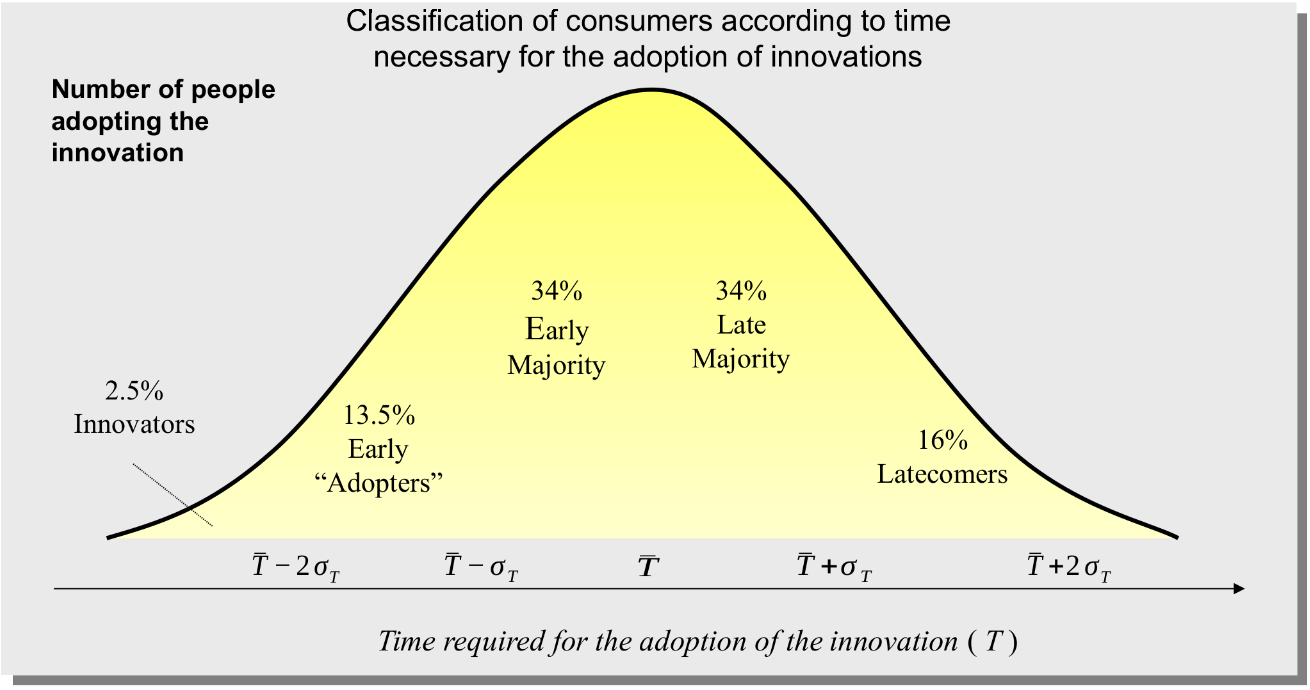
- In the same market, consumers differ in the time they take to try a new product after learning about it

- Early adopters tend to have common characteristics that differentiate them from later adopters

- There are effective ways to reach early adopters of a new product

- Among the early consumers are opinion leaders who can play a useful role by advertising the new product to other potential buyers.

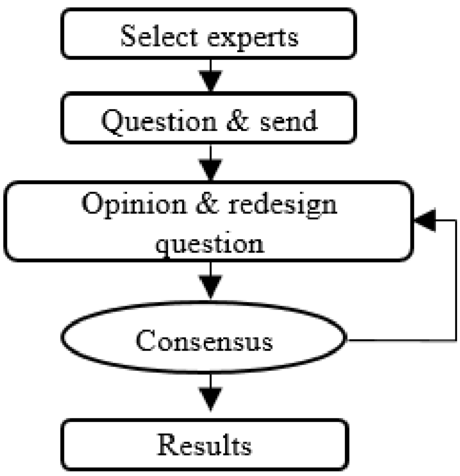
Consumer and Adoption



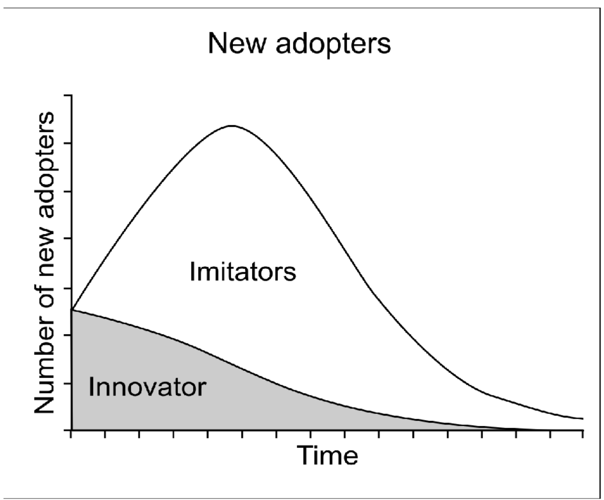
Source : EM Rogers (1983)

How anticipate the diffusion of a new product?

The Delphi method



The Bass Model



The Bass Model

Scientific Marketing

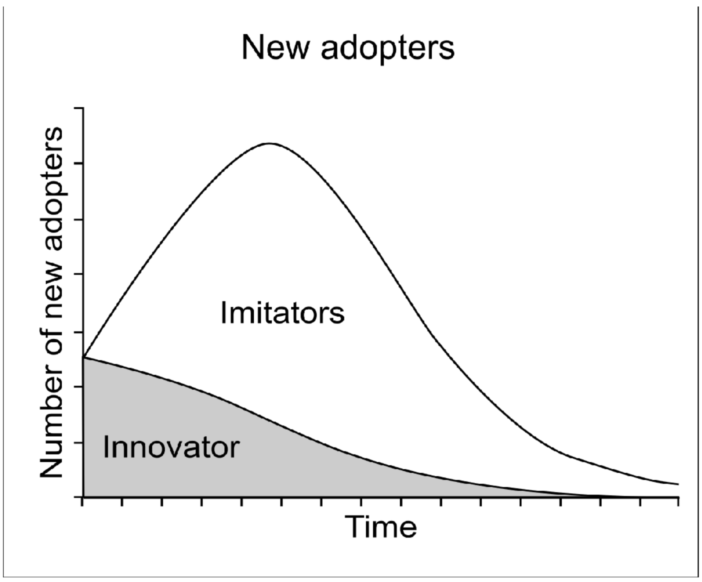
Frank M. Bass (1969), “A New Product Growth Model for Consumer Durables”, *Management Science* , Vol. 15, No. 4 (January), 215-227.

- Model objectives
  - Model to predict the demand for the first equipment of an innovation (**how many** people will adopt the new product and **when** will they adopt it?)
  - Good starting point for forecasting LT sales of durable goods

- Terms of application
  - The innovation has already been introduced to the market such that some sales observations are available
  - Or the innovation has not yet been introduced to the market but sales data for a similar innovation are available

The Bass Model

Consumers

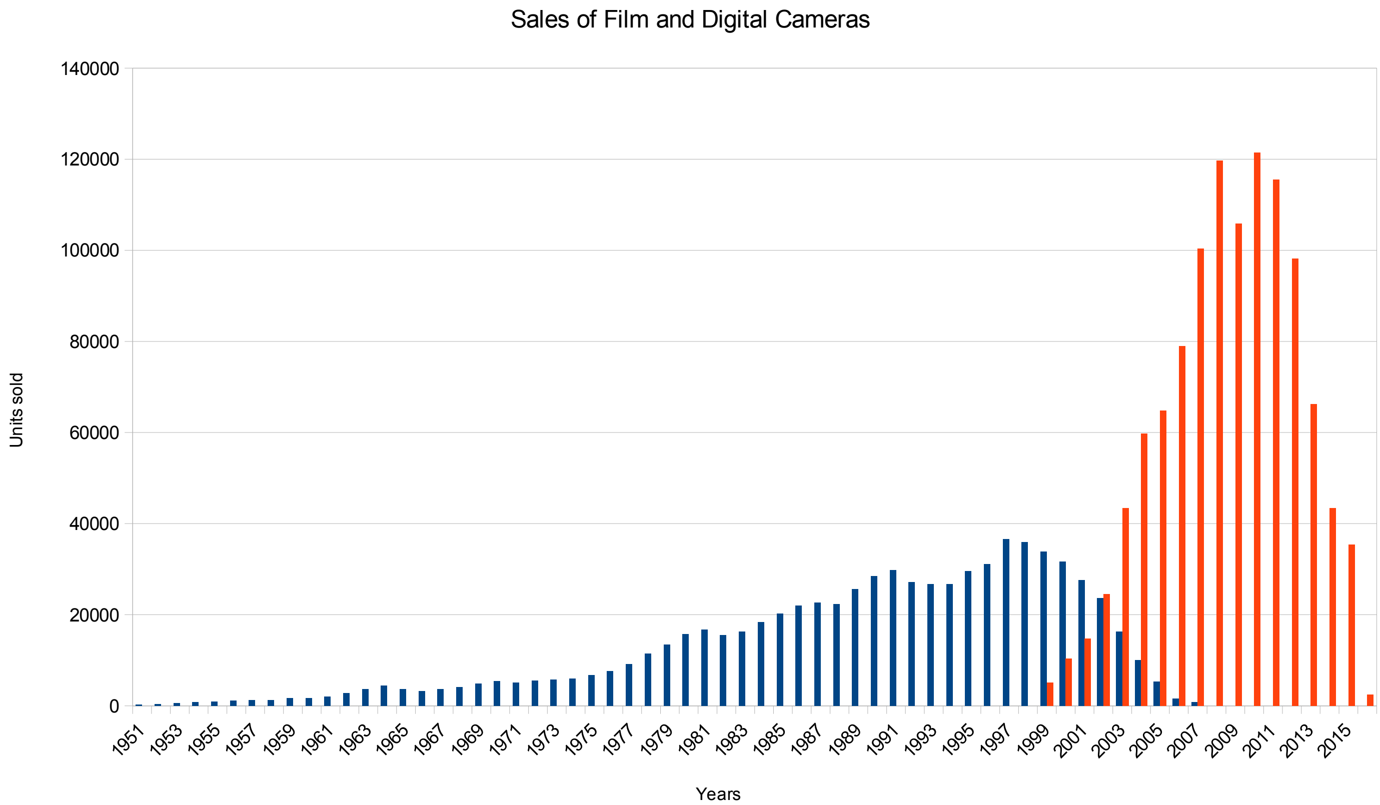


The market is populated by 2 types of consumers

- The innovators (pioneers)
- The imitators

Their numbers change over time

The diffusion of Cameras
The Bass Model



The Bass Model
Model Formulation

- p: innovation coefficient
- q: imitation coefficient
- M: Global potential market
- A_t: Share (%) of consumers who have not yet purchased the product who will buy it at period t

$$A_t = p + q\frac{\sum_{i=1}^{t-1} s_i}{M}$$

- s_i: Sales at period i

Potential for summer market

"Adopters" (%) in t

$$s_t = \left(M - \sum_{i=1}^{t-1} s_i\right) A_t = \left(M - \sum_{i=1}^{t-1} s_i\right)\left(p + q\frac{\sum_{i=1}^{t-1} s_i}{M}\right)$$

$$s_t = pM + (q-p).\sum_{i=1}^{t-1} s_i - \frac{q}{M}.\left(\sum_{i=1}^{t-1} s_i\right)^2$$

The Bass Model

Application: Digital Cameras

$$s_t = pM + (q-p).\sum_{i=1}^{t-1} s_i - \frac{q}{M}.\left(\sum_{i=1}^{t-1} s_i\right)^2$$

$$s_t = \beta_0 + \beta_1.\sum_{i=1}^{t-1} s_i + \beta_2.\left(\sum_{i=1}^{t-1} s_i\right)^2$$

Ventes annuelles des appareils numériques	Cumul des ventes passées	Cumul des ventes passées au carré
5088	0	0
10342	5088	25887744
14753	15430	238084900
24551	30183	911013489
43408	54734	2995810756
59766	98142	9631852164
64766	157908	24934936464
78981	222674	49583710276
100367	301655	90995739025
119757	402022	161621688484

- First, we estimate β $_{0,}$ β $_1$ and β $_2$ using linear regression
- In Excel (or another spreadsheet)
- =LINEST(Variable to be explained ; Explanatory variable(s) ; with or without constant; Coefficients alone or with additional statistics)

The Bass Model

Application: Linear Regression

$$s_t = pM + (q-p).\sum_{i=1}^{t-1} s_i - \frac{q}{M}.\left(\sum_{i=1}^{t-1} s_i\right)^2$$

$$s_t = \beta_0 + \beta_1.\sum_{i=1}^{t-1} s_i + \beta_2.\left(\sum_{i=1}^{t-1} s_i\right)^2$$

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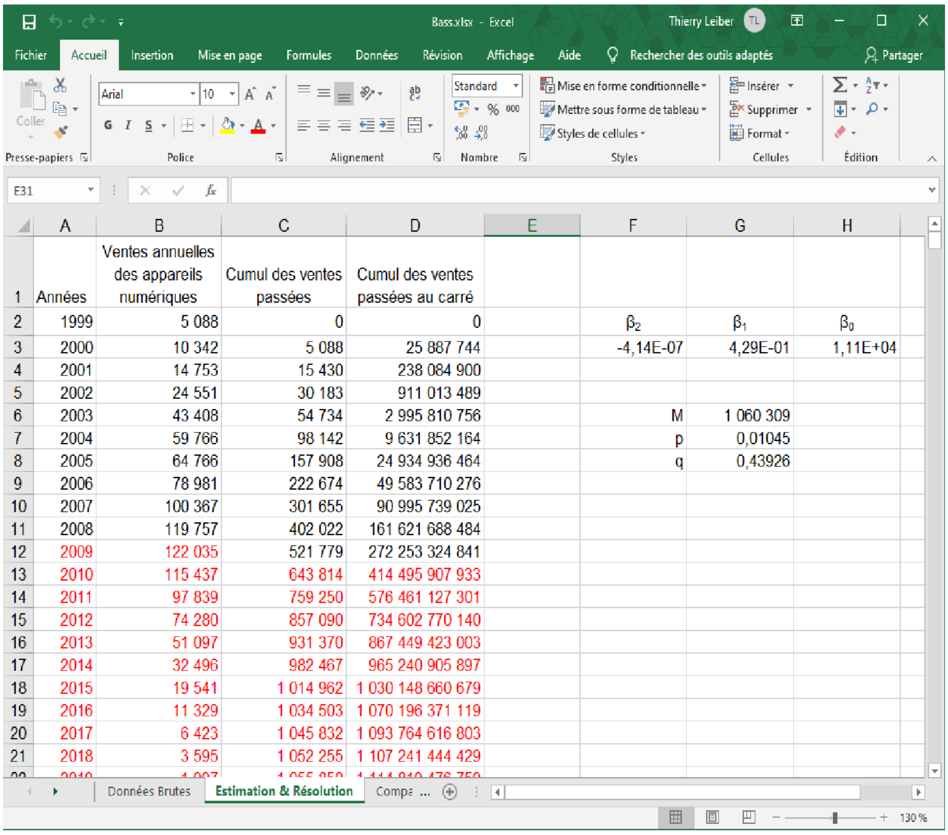
- Linear regression will make it possible to find the coefficients β $_{0,}$ β $_1$ and β $_2$ which make it possible to obtain the best possible approximations of the values v $_t$.
- β $_{0,}$ β $_1$ and β $_2$ such that we have

$$min \sum_{t=1}^{T} \left(s_t - \beta_0 + \beta_1.\sum_{i=1}^{t-1} s_i + \beta_2.\sum_{i=1}^{t-1} s_i^{\,2} \right)^2$$

The Bass Model

Application: Digital Cameras



	A	B	C	D	E	F	G	H
1	Années	Ventes annuelles des appareils numériques	Cumul des ventes passées	Cumul des ventes passées au carré				
2	1999	5 088	0	0		β2	β1	β0
3	2000	10 342	5 088	25 887 744		-4,14E-07	4,29E-01	1,11E+04
4	2001	14 753	15 430	238 084 900				
5	2002	24 551	30 183	911 013 489				
6	2003	43 408	54 734	2 995 810 756		M	1 060 309	
7	2004	59 766	98 142	9 631 852 164		p	0,01045	
8	2005	64 766	157 908	24 934 936 464		q	0,43926	
9	2006	78 981	222 674	49 583 710 276				
10	2007	100 367	301 655	90 995 739 025				
11	2008	119 757	402 022	161 621 688 484				
12	2009	122 035	521 779	272 253 324 841				
13	2010	115 437	643 814	414 495 907 933				
14	2011	97 839	759 250	576 461 127 301				
15	2012	74 280	857 090	734 602 770 140				
16	2013	51 097	931 370	867 449 423 003				
17	2014	32 496	982 467	965 240 905 897				
18	2015	19 541	1 014 962	1 030 148 660 679				
19	2016	11 329	1 034 503	1 070 196 371 119				
20	2017	6 423	1 045 832	1 093 764 616 803				
21	2018	3 595	1 052 255	1 107 241 444 429				

The Bass Model

System Resolution

$$s_t = pM + (q-p).\sum_{i=1}^{t-1} s_i - \frac{q}{M}.\left(\sum_{i=1}^{t-1} s_i\right)^2$$

$$s_t = \beta_0 + \beta_1.\sum_{i=1}^{t-1} s_i + \beta_2.\left(\sum_{i=1}^{t-1} s_i\right)^2$$

β_2	β_1	β_0
-4,14E-07	4,29E-01	1,11E+04

$$\begin{pmatrix} \beta_0 = p.M \\ \beta_1 = q - p \\ \beta_2 = \frac{-q}{M} \end{pmatrix}$$

Isolating p and q and substituting them in the second equation, we get

$$\begin{pmatrix} p = \frac{\beta_0}{M} \\ \beta_1 = -\beta_2.M - \frac{\beta_0}{M} \\ q = -\beta_2.M \end{pmatrix}$$

Passing all sides of the equation to the same side, we get

$$\frac{\beta_0}{M} + \beta_1 + \beta_2.M = 0$$

Multiplying by M to remove the denominator, we finally get

The following quadratic equation

$$\beta_2.M^2 + \beta_1.M + \beta_0 = 0$$

The Bass Model

Forecast made in 2008

β₂	β₁	β₀
-4,14E-07	4,29E-01	1,11E+04

Années	Ventes annuelles des appareils numériques	Cumul des ventes passées	Cumul des ventes passées au carré
1999	5 088	0	0
2000	10 342	5 088	25 887 744
2001	14 753	15 430	238 084 900
2002	24 551	30 183	911 013 489
2003	43 408	54 734	2 995 810 756
2004	59 766	98 142	9 631 852 164
2005	64 766	157 908	24 934 936 464
2006	78 981	222 674	49 583 710 276
2007	100 367	301 655	90 995 739 025
2008	119 757	402 022	161 621 688 484
2009	122 035	521 779	272 253 324 841
2010	115 437	643 814	414 495 907 933
2011	97 839	759 250	576 461 127 301
2012	74 280	857 090	734 602 770 140
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
2025	56	1 060 184	1 123 989 967 157
2026	31	1 060 240	1 124 108 878 167
2027	17	1 060 271	1 124 174 326 539
2028	9	1 060 288	1 124 210 346 166
2029	5	1 060 297	1 124 230 168 715
2030	3	1 060 302	1 124 241 077 310
2031	2	1 060 305	1 124 247 080 360

$$s_t = \beta_0 + \beta_1 . \sum_{i=1}^{t-1} s_i + \beta_2 . \sum_{i=1}^{t-1} s_i^2$$

The Bass Model

System Resolution

β₂	β₁	β₀
-4,14E-07	4,29E-01	1,11E+04

$$\beta_2 . M^2 + \beta_1 . M + \beta_0 = 0$$

Solving the equation gives us 2 values of M, one positive M^+, the other negative M^-

$$M^+ = \frac{-\beta_1 - \sqrt{\beta_1^2 - 4\beta_0 . \beta_2}}{2\beta_2}$$

$$M^- = \frac{-\beta_1 + \sqrt{\beta_1^2 - 4\beta_0 . \beta_2}}{2\beta_2}$$

We keep the single consistent value of M^+ (Global Potential Market) and we calculate p and q

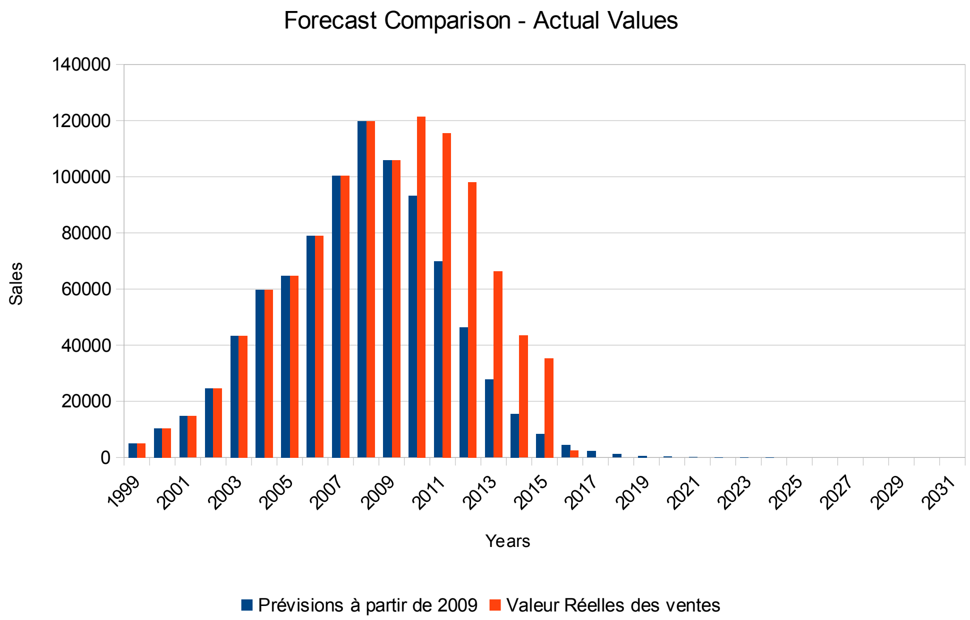
$$\left(\begin{array}{c} M^+ = \frac{-\beta_1 - \sqrt{\beta_1^2 - 4\beta_0 . \beta_2}}{2\beta_2} \\ p = \frac{\beta_0}{M^+} \\ q = -\beta_2 . M^+ \end{array} \right)$$

M	1 060 309
p	0,01045
q	0,43926

The Bass Model

Comparison Forecast – Actual Values

Années	Prévisions à partir de 2009	Valeur Réelles des ventes
1999	5 088	5088
2000	10 342	10342
2001	14 753	14753
2002	24 551	24551
2003	43 408	43408
2004	59 766	59766
2005	64 766	64766
2006	78 981	78981
2007	100 367	100367
2008	119 757	119757
2009	122 035	105864
2010	115 437	121463
2011	97 839	115524
2012	74 280	98139
2013	51 097	66284
2014	32 496	43434
2015	19 541	35395
2016	11 329	2419
2017	6 423	
2018	3 595	
2019	1 997	
2020	1 105	



Use of similar products

- Essential method when the product has not yet been launched or when there is no information on the first sales
  - p and q are estimated using the sales history of an analogous product
  - M is an estimate specific to the innovation (for example: Delphi method)

- But what is an analogous product?
  - Environmental context
  - Market structure
  - Buyer behavior
  - Company marketing strategy
  - Characteristics of innovation (complexity, price,...)

Do yourself a sensitivity analysis

- How do the results change with one year, more or less data ?
- Write your personal conclusion about this technique.
- Try to do the same job with the R Language.